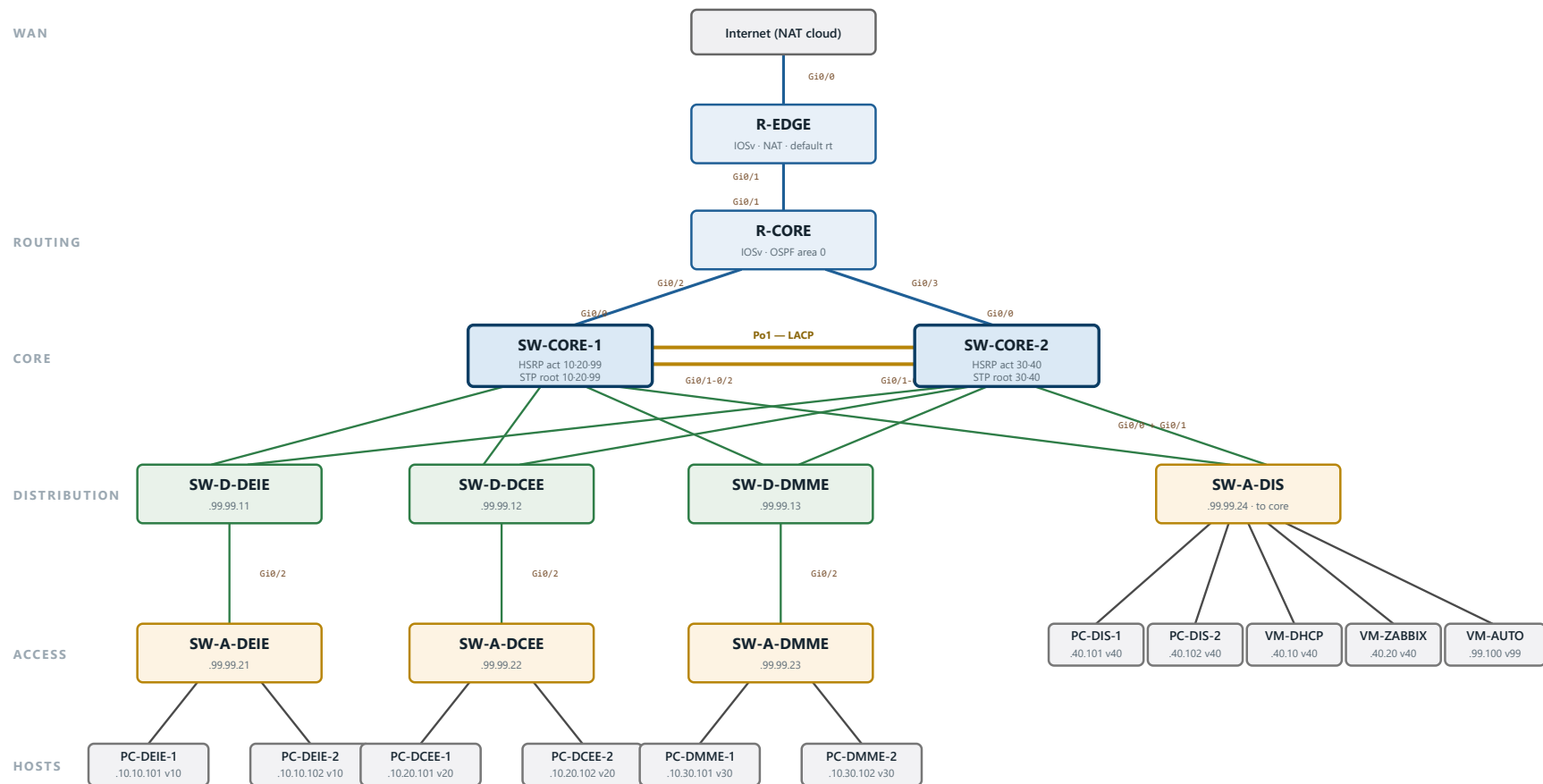


EE8203 — Campus Network Topology

Faculty of Engineering, University of Ruhuna · GNS3 build sheet · v0.1



— Routed link (/30) — 802.1Q trunk — LACP EtherChannel (Po1) — Access port Host addresses abbreviated: .40.101 = 10.10.40.101

Dual-homing is intentional. Each distribution switch and SW-A-DIS has two uplinks — one to each core. This creates a Layer 2 loop by design; spanning tree blocks the redundant path and reactivates it automatically on failure. STP root priority is aligned with the HSRP active role so the unblocked path always leads to the active gateway.

Node list and cabling

Place and rename every node before cabling. Cable in batches A–D and verify each batch.

Step 1 — Place and rename nodes

Qty	GNS3 template	Rename to	RAM each
2	Cisco IOSv 15.9(3)M6	R-EDGE, R-CORE	512 MB
2	Cisco IOSvL2 15.2	SW-CORE-1, SW-CORE-2	768 MB
3	Cisco IOSvL2 15.2	SW-D-DEIE, SW-D-DCEE, SW-D-DMME	768 MB
4	Cisco IOSvL2 15.2	SW-A-DEIE, SW-A-DCEE, SW-A-DMME, SW-A-DIS	768 MB
8	VPCS	PC-DEIE-1/2, PC-DCEE-1/2, PC-DMME-1/2, PC-DIS-1/2	—
1	NAT cloud	Internet	—
3	Ubuntu 22.04 (add later)	VM-AUTO, VM-ZABBIX, VM-DHCP	2 GB / 2.5 GB / 512 MB

Rename each node as you place it. Renaming after cabling is how port maps drift out of sync with the real topology — and the port map is what every later configuration step depends on.

Batch A — Routing spine (4 links)

From	Port	To	Port
Internet	—	R-EDGE	Gi0/0
R-EDGE	Gi0/1	R-CORE	Gi0/1
R-CORE	Gi0/2	SW-CORE-1	Gi0/0
R-CORE	Gi0/3	SW-CORE-2	Gi0/0

Batch B — Core peer link (2 links → Po1)

From	Port	To	Port
SW-CORE-1	Gi0/1	SW-CORE-2	Gi0/1
SW-CORE-1	Gi0/2	SW-CORE-2	Gi0/2

Batch C — Dual-homed uplinks (8 links)

From	Port	To	Port
SW-D-DEIE	Gi0/0	SW-CORE-1	Gi0/3
SW-D-DEIE	Gi0/1	SW-CORE-2	Gi0/3
SW-D-DCEE	Gi0/0	SW-CORE-1	Gi1/0
SW-D-DCEE	Gi0/1	SW-CORE-2	Gi1/0
SW-D-DMME	Gi0/0	SW-CORE-1	Gi1/1
SW-D-DMME	Gi0/1	SW-CORE-2	Gi1/1
SW-A-DIS	Gi0/0	SW-CORE-1	Gi1/2
SW-A-DIS	Gi0/1	SW-CORE-2	Gi1/2

Cabling batch D — access layer and hosts

The final 11 links. After this batch the topology is fully cabled: 25 links in total.

Batch D — Access uplinks and end hosts (11 links)

From	Port	To	Port
SW-A-DEIE	Gi0/0	SW-D-DEIE	Gi0/2
SW-A-DCEE	Gi0/0	SW-D-DCEE	Gi0/2
SW-A-DMME	Gi0/0	SW-D-DMME	Gi0/2

Host	Switch	Port	VLAN
PC-DEIE-1 / PC-DEIE-2	SW-A-DEIE	Gi1/0, Gi1/1	10
PC-DCEE-1 / PC-DCEE-2	SW-A-DCEE	Gi1/0, Gi1/1	20
PC-DMME-1 / PC-DMME-2	SW-A-DMME	Gi1/0, Gi1/1	30
PC-DIS-1 / PC-DIS-2	SW-A-DIS	Gi1/0, Gi1/1	40
VM-DHCP	SW-A-DIS	Gi1/3	40
VM-ZABBIX	SW-A-DIS	Gi2/0	40
VM-AUTO	SW-A-DIS	Gi2/1	99

What to expect once cabling is complete

Normal at this stage

- GNS3 may warn about a Layer 2 loop after batch C — this is by design, ignore it.
- Every interface shows `down/down`. Nothing has been configured yet.
- Switches boot to `Switch>`, routers to `Router>`.
- Nothing pings. No addressing has been applied to any device.
- IOSvL2 takes 2–4 minutes to boot. Wait for the prompt before assuming a fault.

Verify before configuring

- Start only **R-EDGE, R-CORE, SW-CORE-1, SW-CORE-2** (~2.6 GB).
- Each reaches a CLI prompt.
- `show ip interface brief` lists `Gi0/0 – Gi0/3` on routers.
- `show ip interface brief` lists `Gi0/0 – Gi3/3` on switches.
- Save the project, then take a GNS3 snapshot named `00-cabled-baseline`.

Take that snapshot. It is the clean state to roll back to whenever a later configuration step goes wrong, and the MOP requires a documented rollback point before each major change.

VLAN and HSRP reference

The VLAN scheme, and how the redundant core pair presents a single gateway to each subnet.

VLANs

VLAN	Name	Department	Subnet	Gateway (HSRP VIP)	Purpose
10	VLAN_DEIE	DEIE	10.10.10.0/24	10.10.10.1	Engineering workstations
20	VLAN_DCEE	DCEE	10.10.20.0/24	10.10.20.1	Civil / Environmental hosts
30	VLAN_DMME	DMME	10.10.30.0/24	10.10.30.1	Mechanical / Manufacturing hosts
40	VLAN_DIS	DIS	10.10.40.0/24	10.10.40.1	Faculty IT hub / server farm
99	MGMT	All	10.99.99.0/24	10.99.99.1	Device management, SSH only
100	NATIVE	Trunks	none	none	Native (untagged) VLAN on all trunks

HSRP on the core pair

VLAN	SW-CORE-1 SVI	SW-CORE-2 SVI	Virtual IP	Group	Active (pri 110)	STP root
10	10.10.10.2	10.10.10.3	10.10.10.1	10	SW-CORE-1	SW-CORE-1
20	10.10.20.2	10.10.20.3	10.10.20.1	20	SW-CORE-1	SW-CORE-1
30	10.10.30.2	10.10.30.3	10.10.30.1	30	SW-CORE-2	SW-CORE-2
40	10.10.40.2	10.10.40.3	10.10.40.1	40	SW-CORE-2	SW-CORE-2
99	10.99.99.2	10.99.99.3	10.99.99.1	99	SW-CORE-1	SW-CORE-1

Why alternate the active role? Leaving one core as a cold standby wastes half the core capacity. Splitting the VLANs means both switches forward traffic in normal operation, and either one can carry everything after a failure. The standby keeps the default priority 100, and `preempt` is enabled on both so the intended active switch reclaims its role once it recovers.

Why STP root must match HSRP active. If SW-CORE-1 is the gateway for VLAN 10 but SW-CORE-2 is the STP root for it, every VLAN 10 packet reaches CORE-2 at Layer 2, crosses the peer link to find its gateway on CORE-1, and comes back — turning Po1 into a bottleneck for no reason. Aligning the two keeps the forwarding path direct.

Addressing reference

Every address used in the build. Keep this page beside you while configuring.

Point-to-point routed links

Link	Subnet	Side A	Side B
R-EDGE ↔ Internet	203.0.113.0/24	.2 (Gi0/0)	.1 (gw)
R-EDGE ↔ R-CORE	10.255.255.0/30	.1 (Gi0/1)	.2 (Gi0/1)
R-CORE ↔ SW-CORE-1	10.255.255.4/30	.5 (Gi0/2)	.6 (Gi0/0)
R-CORE ↔ SW-CORE-2	10.255.255.8/30	.9 (Gi0/3)	.10 (Gi0/0)

Hosts and servers

Host	Address	VLAN
PC-DEIE-1 / -2	10.10.10.101 / .102	10
PC-DCEE-1 / -2	10.10.20.101 / .102	20
PC-DMME-1 / -2	10.10.30.101 / .102	30
PC-DIS-1 / -2	10.10.40.101 / .102	40
VM-DHCP	10.10.40.10	40
VM-ZABBIX	10.10.40.20	40
VM-AUTO	10.99.99.100	99

Test hosts use static addresses so ACL results are reproducible. DHCP pools serve .150 – .200 in each department VLAN.

Management addresses (VLAN 99)

Device	Address	Interface
SW-CORE-1	10.99.99.2/24	SVI VLAN 99
SW-CORE-2	10.99.99.3/24	SVI VLAN 99
SW-D-DEIE	10.99.99.11/24	SVI VLAN 99
SW-D-DCEE	10.99.99.12/24	SVI VLAN 99
SW-D-DMME	10.99.99.13/24	SVI VLAN 99
SW-A-DEIE	10.99.99.21/24	SVI VLAN 99
SW-A-DCEE	10.99.99.22/24	SVI VLAN 99
SW-A-DMME	10.99.99.23/24	SVI VLAN 99
SW-A-DIS	10.99.99.24/24	SVI VLAN 99
R-CORE	10.99.0.1/32	Loopback0
R-EDGE	10.99.0.2/32	Loopback0

All distribution and access switches use 10.99.99.1 — the VLAN 99 HSRP virtual address — as their default gateway, so management reachability survives the loss of either core switch. Routers sit outside VLAN 99 and are managed on loopbacks advertised into OSPF.

SSH targets for automation

VM-AUTO (10.99.99.100) reaches every device in the management table above. This list becomes the Netmiko inventory file and the Ansible inventory in weeks 5–8.

Memory budget and build checklist

Why you cannot start every device

Group	Devices	RAM
Core and routing	R-EDGE, R-CORE, SW-CORE-1, SW-CORE-2	2.6 GB
Distribution	SW-D-DEIE, SW-D-DCEE, SW-D-DMME	2.3 GB
Access	SW-A-DEIE / DCEE / DMME / DIS	3.1 GB
Servers	VM-AUTO, VM-ZABBIX, VM-DHCP	5.0 GB
GNS3 VM overhead	—	~1.5 GB
Windows host	—	~4.0 GB
Total if everything runs at once		~18.5 GB

The build host has **16 GB**. The topology is never run in full — devices are started per phase as shown opposite.

Staged startup by phase

Phase	Start these	RAM
Baseline routing & trunking	Core group + distribution	9.4 GB
VLANs & host connectivity	+ access switches + VPCS	12.5 GB
ACL policy testing	All network devices + VPCS (no servers)	12.5 GB
Netmiko / Ansible	All network devices + VM-AUTO	14.5 GB
Zabbix monitoring	Core + dist + SW-A-DIS + VM-ZABBIX	11.0 GB

For the current step, start only **R-EDGE, R-CORE, SW-CORE-1, SW-CORE-2** (~2.6 GB). IOSvL2 takes 2–4 minutes to boot — wait for the prompt before assuming a fault.

Build checklist

Placement

- ☐ Project created as `EE8203-campus-network`
- ☐ 2 × IOSv placed and renamed
- ☐ 9 × IOSvL2 placed and renamed
- ☐ 8 × VPCS placed and renamed
- ☐ NAT cloud placed, renamed `Internet`
- ☐ Nodes arranged in tiers (WAN → core → dist → access → hosts)
- ☐ All names checked for typos

Cabling

- ☐ Batch A — routing spine, 4 links
- ☐ Batch B — core peer link, 2 links
- ☐ Batch C — dual-homed uplinks, 8 links
- ☐ Batch D — access and hosts, 11 links
- ☐ Total 25 links, each verified against the port map

Verify before configuring

- ☐ Start R-EDGE, R-CORE, SW-CORE-1, SW-CORE-2
- ☐ Each reaches a CLI prompt
- ☐ `show ip interface brief` lists Gi0/0–Gi0/3 on routers
- ☐ `show ip interface brief` lists Gi0/0–Gi3/3 on switches
- ☐ Project saved
- ☐ GNS3 snapshot taken, named `00-cabled-baseline`

Then, and only then

Move on to device configuration: hostnames, VLAN database, trunk links with native VLAN 100, the core SVIs, HSRP, and management access on VLAN 99. Configure and verify the core pair before starting any distribution or access switch — a fault in the core is far easier to find with four devices running than with eleven.